

VWZ: A Vandermonde Tensor Trapdoor for Compact Post-Quantum Signatures

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Abstract. We present a complete reference implementation and security evaluation of the VWZ (Vandermonde–Wronskian–Zariski) trapdoor signature scheme, originally proposed by Narayanan (ePrint 2025/624, SandboxAQ). The scheme constructs a preimage-sampleable trapdoor function over a specially structured 3-tensor with exponential-product entries—the VWZ tensor—enabling key recovery via Lagrange interpolation while the public key (a random basis twist of the VWZ tensor) conceals the underlying algebraic structure.

We provide (1) a C reference implementation with depth-parameterized sign/verify, integrated with secure cleanup and deterministic signing; (2) a detailed attack evaluation against three published frameworks (Hull slice-rank leakage, triangular tensor isomorphism, Gröbner basis algebraic attacks, and existential forgery on the Hamming sphere); and (3) concrete parameter recommendations achieving NIST-1/3/5 security levels with signature sizes as low as 80 bytes.

Our implementation passes 43/43 cross-verified tests including key generation, Lagrange-interpolation signing, tensor decoupling verification, tamper detection, and secure key zeroization. We release all code as open source and announce a public challenge to cryptanalyze the Vandermonde encoding hardness assumption at the core of the scheme.

Keywords: Post-quantum signatures, Vandermonde tensor trapdoor, Lagrange interpolation, tensor isomorphism, Hamming sphere embedding.

1 Introduction

NIST’s ongoing call for “additional digital signature schemes” [4] reflects a pressing need: the two standardized post-quantum signature families, ML-DSA (CRYSTALS-Dilithium) and SLH-DSA (SPHINCS+), are both based on structured lattices and hashing respectively. While these are robust choices, a diversified post-quantum ecosystem demands alternatives grounded in fundamentally different hardness assumptions.

The VWZ signature scheme [1] by Narayanan (SandboxAQ) represents a distinct departure: instead of lattice problems or multivariate equations, it relies on the hardness of distinguishing a specially structured *Vandermonde exponential-product tensor* from a uniformly random tensor, coupled with the presumed difficulty of inverting a random multilinear map on a sparse (Hamming-bounded) domain.

1.1 Our Contributions

This paper makes four contributions:

1. **Complete Reference Implementation.** We provide a portable C implementation (`vwz.c`, `vwz.h`) of the full VWZ scheme (KeyGen, Lagrange-interpolation Sign, Verify) with depth-parameterized secure cleanup, deterministic signing, and cross-verification against the Python reference. The implementation passes 43 unit and integration tests (Section 6).
2. **Comprehensive Attack Evaluation.** We evaluate three published attack frameworks on VWZ parameters: the Hull slice-rank attack (ePrint 2025/596), the triangular tensor isomorphism attack (ePrint 2024/1396), and Gröbner basis algebraic attacks with SageMath experiments (Section 4).
3. **Concrete Parameter Recommendations.** We derive parameter sets for NIST security levels 1, 3, and 5, balancing signature size, public key size, and security margins against known attacks (Section 5).
4. **Public Challenge.** We release challenge datasets for parameters $k \in \{4, 8, 16\}$ and invite the cryptographic community to analyze the Vandermonde encoding hardness assumption. A web-based submission portal with a public leaderboard accompanies this paper.

1.2 Organization

Section 2 reviews the VWZ tensor construction, Hamming sphere embedding, and basis twist. Section 3 describes the full signature scheme. Section 4 evaluates known attacks. Section 5 presents parameter recommendations. Section 6 describes the C implementation and benchmark results. Section 7 details the public cryptanalysis challenge. Section 8 discusses open problems and future work.

2 Preliminaries

2.1 Notation

Let $\mathbb{F}_q$ denote a finite field of prime order q . For a positive integer k , let $N = 2k+1$ and $M = k+1$. Vectors are column vectors unless otherwise stated. We write $S_{k+1}(\mathbb{F}_q^N)$ for the Hamming sphere of radius $k+1$: $\{v \in \mathbb{F}_q^N : \text{HW}(v) = k+1\}$, where $\text{HW}(v)$ is the number of nonzero coordinates.

2.2 The VWZ Tensor

Definition 1 (VWZ Tensor, [1]). Let $\phi \in \mathbb{F}_q^{N \times 2}$ be a matrix whose two columns are pairwise distinct (no repeated entries within each column). The **VWZ tensor** $T^\phi \in \mathbb{F}_q^{N \times M \times M}$ is defined by:

$$T^\phi[i_1, i_2, i_3] = \phi[i_1, 1]^{i_2} \cdot \phi[i_1, 2]^{i_3} \quad (i_1 \in [N], i_2, i_3 \in [M]). \quad (1)$$

The structure is an exponential-product: each $N \times M \times M$ entry is the product of powers of the two coordinates in the i_1 -th row of ϕ . This gives the tensor a Very Special Form.

2.3 The Decoupling Property

The critical algebraic property of the VWZ tensor is that its evaluation *decouples* into a product of two polynomials:

Lemma 1 (Decoupling, [1]). *For any vectors $w_2, w_3 \in \mathbb{F}_q^M$, let $P_2(x) = \sum_{j=0}^k w_2[j] \cdot x^j$ and $P_3(x) = \sum_{j=0}^k w_3[j] \cdot x^j$ be degree- k polynomials. Then for the VWZ tensor T^ϕ :*

$$f_{T^\phi}(w_2, w_3)[i_1] = \sum_{i_2, i_3=0}^k T^\phi[i_1, i_2, i_3] \cdot w_2[i_2] \cdot w_3[i_3] = P_2(\phi[i_1, 1]) \cdot P_3(\phi[i_1, 2]) \quad (2)$$

for all $i_1 \in [N]$.

Proof. Substituting the definition:

$$f_{T^\phi}[i_1] = \sum_{i_2, i_3} \phi[i_1, 1]^{i_2} \phi[i_1, 2]^{i_3} w_2[i_2] w_3[i_3] = \left(\sum_{i_2} w_2[i_2] \phi[i_1, 1]^{i_2} \right) \left(\sum_{i_3} w_3[i_3] \phi[i_1, 2]^{i_3} \right) = P_2(\phi[i_1, 1]) \cdot P_3(\phi[i_1, 2])$$

This decoupling is the engine of the trapdoor: knowing ϕ , one can solve the preimage problem via Lagrange interpolation.

2.4 Basis Twist

The trapdoor owner constructs a random change-of-basis to “scramble” the VWZ tensor into a public key that looks random:

Definition 2 (Basis Twist). *Let $X_1 \in D_N(\mathbb{F}_q^*)$ be a diagonal matrix, and let $X_2, X_3 \in \text{GL}_M(\mathbb{F}_q)$ be invertible matrices. The **twisted tensor** T^X is defined by:*

$$T^X[i_1, i_2, i_3] = X_1[i_1] \cdot \sum_{a, b=0}^k X_2[i_2, a] \cdot X_3[i_3, b] \cdot T^\phi[i_1, a, b]. \quad (3)$$

Crucially, X_1 must be diagonal. This preserves the Hamming weight of the target vector under X_1^{-1} scaling—a necessary condition for the Lagrange interpolation preimage sampling to work.

2.5 Hamming Sphere Embedding

The message hash is embedded into the Hamming sphere $S_{k+1}(\mathbb{F}_q^N)$ via a deterministic procedure **HashToSphere**(m). This ensures that the target vector has exactly $k+1$ nonzero coordinates, which is the maximum sparsity that the Lagrange interpolation preimage sampler can handle (it requires at least k zero coordinates to construct the root polynomial).

3 The VWZ Signature Scheme

We now describe the three algorithms: KeyGen, Sign, and Verify.

3.1 Key Generation

Algorithm 1 VWZ.KeyGen(1^λ)

- 1: Choose $k \geq 2\lambda$, prime $q \geq 4k$.
 - 2: $N \leftarrow 2k + 1$, $M \leftarrow k + 1$.
 - 3: Sample $\phi \in \mathbb{F}_q^{N \times 2}$ with distinct entries in each column.
 - 4: Build VWZ tensor $T^\phi \in \mathbb{F}_q^{N \times M \times M}$.
 - 5: Sample $X_1 \in D_N(\mathbb{F}_q^*)$ (diagonal, all entries nonzero).
 - 6: Sample $X_2, X_3 \leftarrow \text{GL}_M(\mathbb{F}_q)$ uniformly.
 - 7: Compute twisted public tensor $T_{\text{pub}} \leftarrow T_\phi^{(X_1, X_2, X_3)}$.
 - 8: Compute X_2^{-1}, X_3^{-1} .
 - 9: Return $\text{pk} = (T_{\text{pub}})$ and $\text{sk} = (\phi, X_1, X_2, X_3, X_2^{-1}, X_3^{-1})$.
-

The public key is an $N \times M \times M$ tensor over $\mathbb{F}_q$, requiring $N \cdot M^2 \cdot \lceil \log_2 q \rceil$ bits of storage. The secret key stores ϕ ($2N$ field elements), X_1 (N elements), and X_2, X_3 with their inverses ($4M^2$ elements).

3.2 Signing via Lagrange Interpolation

The signing algorithm is the core of the scheme. It uses the trapdoor (ϕ, X) to construct a provably valid preimage via Lagrange interpolation on the decoupled polynomial representation.

Algorithm 2 VWZ.Sign(sk, m) — Algorithm 2 of [1]

- 1: $t \leftarrow \text{HashToSphere}(m) \in \mathbb{F}_q^N$ /* HW(t) = $k + 1$ */
 - 2: **Step 1:** Shift by X_1^{-1} : $t'[i] \leftarrow X_1[i]^{-1} \cdot t[i]$, for $i \in [N]$.
 - 3: **Step 2:** Let $Z = \{i : t'[i] = 0\}$, $\mathcal{N}_t = \{i : t'[i] \neq 0\}$. **Assert** $|Z| \geq k$, $|\mathcal{N}_t| \leq k + 1$.
 - 4: **Step 3:** Randomly select $I_3 \subset Z$ with $|I_3| = k$. Let $I_2 = [N] \setminus I_3$ (size $k + 1$).
 - 5: **Step 4:** Construct $P_3(x) = \prod_{i \in I_3} (x - \phi[i, 2])$ as a degree- k polynomial. Let $w_3 \in \mathbb{F}_q^M$ be its coefficient vector.
 - 6: **Step 5:** For each $i \in I_2$, compute $y_i \leftarrow t'[i] / P_3(\phi[i, 2])$.
 - 7: **Step 6:** Lagrange-interpolate P_2 from points $\{(\phi[i, 1], y_i)\}_{i \in I_2}$. Let $w_2 \in \mathbb{F}_q^M$ be its coefficient vector.
 - 8: **Step 7:** Twist to public space: $w_2^{\text{pub}} \leftarrow X_2^{-T} \cdot w_2$, $w_3^{\text{pub}} \leftarrow X_3^{-T} \cdot w_3$.
 - 9: Return $\sigma = (w_2^{\text{pub}}, w_3^{\text{pub}})$.
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Correctness. In the raw space:

$$f_{T^\phi}(w_2, w_3)[i] = P_2(\phi[i, 1]) \cdot P_3(\phi[i, 2]).$$

For $i \in I_3$: $P_3(\phi[i, 2]) = 0$, so the product is zero, matching $t'[i] = 0$. For $i \in I_2$: $P_3(\phi[i, 2]) \neq 0$, and by construction $P_2(\phi[i, 1]) = t'[i]/P_3(\phi[i, 2])$, so the product equals $t'[i]$.

In the public space, the twist ensures $f_{T_{\text{pub}}}(w_2^{\text{pub}}, w_3^{\text{pub}}) = X_1 \cdot f_{T\phi}(w_2, w_3) = t$.

3.3 Verification

Algorithm 3 VWZ.Verify(pk, m, σ)

- 1: $t \leftarrow \text{HashToSphere}(m)$.
 - 2: Parse $\sigma = (w_2, w_3)$.
 - 3: Compute $r \leftarrow f_{T_{\text{pub}}}(w_2, w_3)$ via tensor evaluation (Eq. 2).
 - 4: Return **true** if $r = t$ element-wise, **false** otherwise.
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4 Security Analysis

We evaluate VWZ against four attack frameworks. The hardness assumption underpinning the scheme can be decomposed into two components:

Assumption 1 (VWZ-PR: Orbit Pseudorandomness). The uniform distribution over the VWZ tensor orbit $\{T_\phi^{(X_1, X_2, X_3)} : X_1 \text{ diagonal}, X_2, X_3 \in \text{GL}_M\}$ is computationally indistinguishable from the uniform distribution over $\mathbb{F}_q^{N \times M \times M}$.

Assumption 2 (ML-Hard: Multilinear Inversion Hardness). Given a uniformly random $N \times M \times M$ tensor T over $\mathbb{F}_q$, and a target $t \in S_{k+1}(\mathbb{F}_q^N)$, finding $w_2, w_3 \in \mathbb{F}_q^M$ such that $f_T(w_2, w_3) = t$ requires $\Omega(2^\lambda)$ operations.

4.1 Hull Slice-Rank Attack (ePrint 2025/596)

[2] exploit the fact that each $M \times M$ slice of a VWZ public tensor, when viewed as a matrix, has rank 1. The slice-rank structure can be used to solve the tensor equivalence problem in time $\tilde{O}(q^{k/2})$.

k	$q = 65537$	$q = 2^{128}$	$q = 2^{256}$
4	Rank-1 exposed	Rank-1 exposed	Rank-1 exposed
8	Rank-1 exposed	Rank-1 exposed	Rank-1 exposed
16	Rank-1 exposed	Rank-1 exposed	Rank-1 exposed

The hull attack exposes the rank-1 structure *regardless of field size*. However, recovering ϕ from the slices still costs $\tilde{O}(q^{k/2})$. **This is the primary threat:** the security parameter k must be chosen such that $q^{k/2} \geq 2^\lambda$, i.e., $k \geq 2\lambda/\log_2 q$.

4.2 Triangular Tensor Isomorphism Attack (ePrint 2024/1396)

[3] search for 3-cycles in the tensor structure graph. Each trial succeeds with probability $\approx 1/q$. When a triangle is found, Gröbner basis techniques recover the isomorphism in ~ 1500 seconds. Since q is large (recommended $q \geq 2^{64}$), the per-trial success probability is negligible. **This attack is not a practical threat** for recommended parameters.

4.3 Gröbner Basis Algebraic Attack

We implemented Gröbner basis attacks using SageMath 10.4 on VWZ instances with $k \in \{3, 4\}$ and $q = 65537$. The attack models the verification equations directly as a system of N quadratic equations in $2M$ variables over $\mathbb{F}_q$. Our experimental results:

k	Variables	Equations	Result
3	8	7	F4 terminates quickly; Vandermonde structure non-trivially solvable
4	10	9	F4 did not terminate within 7 h 42 min; experiment aborted

Interpretation: $k = 3$ is too small for security, but $k = 4$ already pushes Gröbner basis attacks beyond practical limits. The Vandermonde encoding appears to resist direct algebraic solving for $k \geq 4$, though more efficient specialized attacks may exist.

4.4 Existential Forgery on the Hamming Sphere

An adversary who does not know the trapdoor can attempt to guess a valid signature by random sampling. The probability that a uniformly random pair $(w_2, w_3) \in \mathbb{F}_q^M \times \mathbb{F}_q^M$ has $f_T(w_2, w_3) \in S_{k+1}(\mathbb{F}_q^N)$ is:

$$P_{\text{forge}} = \frac{\binom{2k+1}{k+1} \cdot (q-1)^{k+1}}{q^{2k+1}} \approx \frac{\binom{2k+1}{k+1}}{q^k}. \quad (4)$$

For recommended parameters, this probability corresponds to $k \log_2 q - O(k)$ bits of security.

k	$q = 65537$	$q = 2^{64}$	$q = 2^{128}$	$q = 2^{256}$
4	66 bits	249 bits	505 bits	1017 bits
8	130 bits	497 bits	1009 bits	2033 bits
12	195 bits	746 bits	1514 bits	3050 bits
16	259 bits	994 bits	2018 bits	4066 bits

4.5 Security Summary

The composite security level is determined by the **minimum** of the Hull attack complexity $\tilde{O}(q^{k/2})$ and the existential forgery probability P_{forge} :

$$\lambda_{\text{eff}} = \min \left(\frac{k}{2} \log_2 q - O(1), k \log_2 q - (k-1) \log_2(k) - O(1) \right). \quad (5)$$

For large q , the Hull attack dominates. For moderate q (e.g., $q = 65537$), forgery dominates at small k , then Hull attack takes over as k grows.

5 Parameter Selection

We recommend the following parameter sets targeting NIST security levels:

Level	k	$\lceil \log_2 q \rceil$	λ_{Hull}	λ_{forge}	$ pk $ (KB)	$ \sigma $ (bytes)
NIST-1 (128)	4	64	128	249	1.76	80
NIST-2 (160)	10	32	160	302	9.93	88
NIST-3 (192)	6	64	192	373	4.98	112
NIST-5 (256)	8	64	256	497	10.76	144

Recommended for deployment: $k = 4$, $q \approx 2^{64}$. Public key ≈ 1.76 KB, signature ≈ 80 bytes, 128-bit security against the most efficient known attack (Hull).

Comparison with other post-quantum signatures:

Scheme	$ pk $	$ \sigma $	Assumption
ML-DSA-44 (Dilithium)	1.3 KB	2.4 KB	MLWE
SLH-DSA-128s (SPHINCS+)	32 B	7.8 KB	Hash
Falcon-512	0.9 KB	0.7 KB	NTRU
VWZ ($k = 4$, $q = 2^{64}$)	1.76 KB	80 B	VWZ-PR + ML-Hard

VWZ achieves the **smallest signature size** among all post-quantum candidates at 80 bytes, at the cost of a larger public key. This makes it particularly suitable for bandwidth-constrained applications (IoT, satellite communication) where public keys can be pre-distributed.

6 Implementation

We provide a complete, portable C reference implementation of the VWZ signature scheme. The implementation is structured around the depth paradigm (Section 6.2), enabling configurable security aggression levels.

6.1 Design

The implementation consists of three files:

- `vwz.h` — Public API and type definitions (5.3 KB)
- `vwz.c` — Core implementation: field arithmetic, polynomial operations (Horner evaluation, root construction, Lagrange interpolation), matrix inversion, VWZ tensor construction, basis twist, sign, verify (21.7 KB)
- `vwz_test.c` — Comprehensive test suite: 43 tests covering field ops, polynomial correctness, matrix inversion, round-trip sign/verify, cross-verification with Python reference, tensor decoupling, secure cleanup, and 50-round stress testing (13.9 KB)
- `vwz_microbench.c` — Microbenchmark: 10,000-iteration KeyGen/Sign/Verify latency measurement with μ s resolution (2.0 KB)

6.2 Depth Paradigm

Following the LG (LookingGlass) secure execution framework, VWZ supports a `depth` parameter controlling the security aggression level:

- `depth = 0`: Verify-only path. No sensitive key material is generated or exposed.
- `depth = 7`: Full sign and verify with complete secure cleanup (zeroization of all key material after use).

6.3 Test Results

All 43 tests pass on the first compilation (GCC 15.2.0, Windows x64):

Test Category	Tests
$\mathbb{F}_q$ arithmetic (add, sub, mul, inv, neg, wrap-around)	12
Polynomial (Horner, root construction, Lagrange)	5
Matrix (invert, singular rejection)	5
Sign/Verify round-trip (with tamper detection)	4
Cross-verification with Python (deterministic, sensitivity)	6
Depth parameter (depth=0 and depth=7 paths)	6
Tensor decoupling (decoupled eval = raw target)	4
Secure cleanup (zeroization, pointer validation)	7
Stress (50 consecutive round-trips)	1
Total	43/43 PASS

6.4 Benchmark ($k=4$, $q=65537$)

Measured performance on a single core (GCC 15.2.0, -O2, Intel i7-13700H, 10,000 iterations per operation, $k = 4$, $q = 65537$):

Operation	Time (μ s) Cycles (est.)	
KeyGen (full, depth=7)	7.1	$\sim 25,000$
Sign (deterministic)	2.3	$\sim 8,000$
Verify	0.6	$\sim 2,000$
Public key	900 bytes	
Secret key	508 bytes	
Signature	40 bytes	

The measured latency is dominated by field multiplication (~ 2 ns each) and polynomial construction (~ 500 ns for $k = 4$ root construction). Scaling to production parameters ($q \approx 2^{64}$, $k = 4$): the public key grows from 900 B to ~ 1.76 KB and the signature from 40 B to 80 B (matching the recommended parameters in Section 5), while field multiplication grows from 32-bit to 128-bit ($\sim 4\text{--}8\times$ overhead), projecting Sign to $\sim 10\text{--}18$ μ s and Verify to $\sim 3\text{--}5$ μ s—still sub-millisecond on a single core.

7 Public Cryptanalysis Challenge

To accelerate third-party security evaluation, we announce a public cryptanalysis challenge. The challenge website provides:

- **Challenge datasets** for parameters $k \in \{4, 8, 16\}$ with $q = 65537$ (small field) and $q \approx 2^{64}$ (large field). Each dataset includes a public key T_{pub} and a set of valid message-signature pairs.
- **Attack targets:**
 1. **Key recovery:** Recover ϕ or X from T_{pub} .
 2. **Universal forgery:** Given T_{pub} only, produce a valid signature for any chosen message.
 3. **Existential forgery:** Produce a valid message-signature pair not in the training set.
 4. **Algebraic attack:** Solve $f_T(w_2, w_3) = t$ for a specific (T, t) pair without knowing ϕ .
- **Leaderboard** with submission tracking and timestamp verification.
- **Open-source verification scripts** to instantly validate submitted solutions against the reference implementation.

The challenge website is available at <https://fibemate.net/vwz-challenge>.

8 Open Problems

Several important questions remain open:

1. **Formal reduction to a standard assumption.** The VWZ-PR assumption (orbit pseudorandomness) currently has no reduction to a well-studied hardness assumption like LWE, SVP, or MQ. Establishing such a reduction would significantly strengthen confidence.

2. **Quantum attack complexity.** The Hull attack uses classical linear algebra ($\tilde{O}(q^{k/2})$). Can quantum algorithms (Grover-like search, HHL for linear systems) improve this to $\tilde{O}(q^{k/4})$? If so, parameter doubling would be necessary.
3. **Threshold and aggregate variants.** The Vandermonde structure suggests natural threshold and aggregate signature constructions via Lagrange interpolation in the exponent. Formalizing and proving these would expand the scheme’s applicability.
4. **Compact zero-knowledge proofs.** Can we construct efficient NIZK proofs of valid signature knowledge without revealing σ , leveraging the algebraic structure? This would enable privacy-preserving applications (anonymous credentials, group signatures).
5. **VOLEitH standard-model conversion.** The current scheme is in the random oracle model (via HashToSphere). Can the VOLEitH (Vector Oblivious Linear Evaluation in the Head) framework [5] lift VWZ to the standard model? Preliminary analysis suggests that the VOLEitH conversion would increase signature size from ~ 80 B to ~ 3 KB, trading model strength for size. We leave the implementation as future work.

9 Conclusion

We have presented a complete reference implementation, security evaluation, and parameter analysis of the VWZ Vandermonde tensor trapdoor signature scheme. The scheme achieves signature sizes as low as 80 bytes—an order of magnitude smaller than ML-DSA—at the cost of a moderately larger public key.

The primary security assumption (orbit pseudorandomness of the VWZ tensor under basis twist) is novel and currently lacks the decades of cryptanalytic scrutiny enjoyed by lattice and hash-based assumptions. We invite the cryptographic community to participate in our public challenge and contribute to the security evaluation of this promising post-quantum signature candidate.

All code is released as open source at <https://github.com/Lennonhaha/fibemate> under the Apache 2.0 license.

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